

CICIDS2017 vs UNSW-NB15 — Dataset Difficulty Comparison

Metric	UNSW-NB15	CICIDS2017
Accuracy	42.96%	56.93
Precision	40.52%	6.83%
Recall	7.69%	4.59%
F1 score	12.92%	7.21
False positive	13.82%	13.02%

Which Dataset Was Harder?

CICIDS2017 was harder for the Isolation Forest model in this experiment.

CICIDS2017 had a lower recall of **4.59%** compared with **7.69%** on UNSW-NB15. Its F1-score was also lower at **7.21%**, compared with **12.92%** for UNSW-NB15.

Although CICIDS2017 had higher accuracy (**56.93%** vs **42.96%**), accuracy alone does not show how well the detector identifies attacks. Recall and F1-score are more useful for this comparison because they show how effectively the model detects anomalous traffic.

Reason:

The Isolation Forest struggled more to identify attacks in CICIDS2017 using the selected six network features. The model detected only **2,318 of 50,535 attack samples** in its test set, resulting in a low recall.

Therefore, for this particular experiment and model configuration, CICIDS2017 was the more difficult dataset.